Rahul Rawat

Working Student, Engagement Lead for AI Vision Cases

PERSONAL DETAILS

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Date of birth: 20 February 2002
Nationality: Indian, student visa with valid work permit
Availability: Werkstudent 20 hours per week immediately, full time from April 2027

PROFILE

Master's student in Data Science and Analytics at SRH Heidelberg, based in Mannheim, with hands on experience in AI systems, LLM integration, and evaluation harnesses. I built a multi agent RAG system with an LLM as Judge evaluation running locally on Ollama with Mistral 7B and Qwen2.5 14B, and delivered a recursive time series pipeline for four water quality indicators at eRay GmbH using CatBoost MultiQuantile with 80 percent prediction intervals. Comfortable across Python, SQL, scikit-learn, LangGraph, and stakeholder communication.

SKILLS

Python, SQL, PySpark, BigQuery, Databricks, Delta Lake, Power BI, Tableau, Looker Studio, LangChain, LangGraph, Ollama, CatBoost, LightGBM, XGBoost, Prophet, MICE, scikit-learn, PyTorch, dbt, Apache Airflow, GCP, AWS, Docker, Git, React, Playwright

PROFESSIONAL EXPERIENCE

Data Scientist at eRay GmbH, Heidelberg, Oct 2025 to Mar 2026

- During a **6 month** eRay GmbH and SRH Heidelberg collaboration to forecast lake water quality across **4 target** indicators chlorophyll a, turbidity, pH and dissolved oxygen, built an end to end recursive time series pipeline over a **40 feature** space with a per target lag suite lag_1h, lag_24h, lag_3d, lag_7d, lag_roll_mean_24h and lag_roll_std_24h.
- Benchmarked **6 candidates** Ridge, Gradient Boosting, LightGBM, XGBoost, CatBoost and Prophet with strict tree constraints max_depth 4 and learning_rate **0.05**, landed on CatBoost MultiQuantile at alpha **0.05**, **0.5** and **0.85**, producing asymmetric **80 percent** prediction intervals that hug the 0 floor and chop the top **15 percent** of summer ghost spikes.
- Made the September evaluation defensible with a **3 pass** outlier system pH tightened from **0 to 14** down to **7.0 to 9.0**, Oct and Nov caps of **15.0** on chlorophyll a and **50.0** on turbidity, and a rolling z-score at z>**2.5** over **48 hours**, and excluded 5 sparse sensors plus 3 concurrent proxies phycocyanin_abs, phycocyanin_abs_comp and toc, surfacing the honest R squared of **0.86** on dissolved oxygen and **0.81** on pH.
- Reconstructed Oct and Nov gaps with IterativeImputer MICE, ran full Memory Buffer recalculation across all 6 lag features, generated a synthetic winter canvas with 4 degree Celsius floor and **0.4** degree diurnal amplitude, then wrapped it all in an orchestrator with gate checks, ecological clips dissolved oxygen **4.0 to 18.0** and pH **6.0 to 9.0** and a **0.003** pH per hour velocity clamp.

Technologies: *Python, CatBoost, Prophet, scikit-learn, MICE*

Junior Associate Software Developer at SS Engineers and Contractors, India, Aug 2023 to Aug 2024

- With SS Engineers and Contractors running internal Data Dashboards, Analytics platforms, and Employee Portals used across the company, tasked with building and maintaining the front end features that day to day teams depended on, contributed React UI components across all three internal products, which stayed in daily use by internal teams across the company throughout the year in role.
- With a client relying on a legacy AngularJS app that had to sit inside an existing module federation setup alongside newer React micro frontends, tasked with porting the client's screens across without breaking the running product, ported around **8 routes** from AngularJS to React inside the module federation shell over **4 months** of incremental releases, shipped with no production incidents during rollout.

Technologies: *React, module federation, Playwright, AngularJS, HTML5, CSS3*

Front End Developer Intern at SS Engineers and Contractors, India, Feb 2023 to July 2023

- During a six month internship at SS Engineers and Contractors, tasked with learning the codebase and then taking on small UI work across the internal Data Dashboards and Employee Portals under senior review, paired closely with senior developers to walk the codebase and then shipped focused UI components including charts, filters, and profile pages, iterating each one on code review feedback until it landed.

Technologies: *React, HTML5, CSS3, Git*

EDUCATION

M.Sc. Data Science and Analytics, SRH University of Applied Sciences Heidelberg, GPA 1.9, Apr 2025 to Present

Bachelor of Technology, GL Bajaj Institute of Technology and Management, CGPA 7.3 of 10, 2019 to 2023

PERSONAL PROJECTS

Multi-Agent RAG with LLM-as-Judge and Multilingual EN and DE Support, built with Python, LangGraph multi agent, Ollama with Mistral 7B and Qwen2.5 14B, Pinecone, paraphrase multilingual MiniLM L12 v2, spaCy, BM25

- For an English only hybrid BM25 plus dense RAG policy analysis system over a **14 document** policy corpus that could not serve German speakers, tasked with adding multilingual support without duplicating the corpus, migrated embeddings and retrieval to a paraphrase multilingual MiniLM L12 v2 shared vector space so a German query retrieves English sources and is answered in German end to end.
- With every agent in the graph re running language detection and producing inconsistent output languages, tasked with a single source of truth, built a LanguageAgent that centralises seeded language detection with a confidence floor, propagates the output language directive to every downstream agent, and routes preprocessing to language matched spaCy pipelines with a blank multilingual fallback and per chunk language recording so retrieval and evaluation can slice by language.
- To measure answer quality without hand grading, implemented an LLM as Judge JudgeAgent scoring answers **1 to 5** on **5 dimensions** (groundedness, relevance, completeness, citation quality, language quality) in JSON mode at temperature 0, and eliminated self preference bias by running the judge on a different local model Qwen**2.5 14B** from the generator Mistral **7B** with a self_judged flag propagated into every report and a hard failure on missing judge model so a silent fallback to self judging cannot regress unnoticed.

CreditIQ: Fairness by Design Credit Scoring, built with Python, scikit learn, AIF360, SHAP, Streamlit

- For an SRH Heidelberg project on regulated credit scoring where a baseline model was failing the EU AI Act and AGG **80 percent** fairness bar, tasked with getting it back into compliance without gutting predictive quality, applied AIF360 mitigation and threshold calibration on a real credit dataset, which raised the Disparate Impact ratio from a failing **0.79** to a compliant **0.88**.
- After single axis age bias was fixed but younger women were still being penalised, tasked with finding and correcting the hidden intersectional bias, used SHAP driven subgroup analysis to expose the pattern and designed a four way age by gender threshold matrix, which corrected it without over correcting into reverse discrimination.
- With a large false negative rate silently rejecting good applicants, tasked with cutting it while keeping overall accuracy defensible, documented the fairness accuracy trade off as a deliberate and regulator defensible decision, which brought the false negative rate down from **44 percent to 16.7 percent** while accuracy held at **75 percent** on the held out test split.

RESEARCH AND THESIS

Bachelor Thesis, Diabetes Prediction Using Machine Learning, GL Bajaj Institute of Technology and Management, 2022 to 2023

- For a Bachelor thesis on diabetes prediction with a small clinical dataset of **768 patients**, tasked with building a defensible model comparison the examiners could audit, built a full end to end machine learning pipeline comparing six classifiers with **10 fold** cross validation and per model confusion matrices, delivering a model comparison that stood up in the thesis defence.
- Spotting biologically impossible zero values in the source data that the original authors had overlooked, tasked with restoring data integrity before any model fit, applied IQR based outlier removal and proper imputation, which lifted the dataset from silently broken to a clean training input for every downstream model.
- With a **65 to 35 class** imbalance that made accuracy a misleading headline metric, tasked with choosing an evaluation that would not hide errors on the minority class, moved the headline metric from accuracy to ROC AUC, which exposed the real error patterns the accuracy score had been masking and gave the thesis an honest performance comparison.
- With the results needing to be publishable in substance rather than just submissible, tasked with writing them up formally, produced an IEEE style paper including an honest limitations section and what to do differently in a follow up study, which the supervisor accepted as publishable in substance.

Technologies: *Python, scikit-learn, Pandas, Seaborn, Google Colab*

CERTIFICATIONS

- NVIDIA: Building LLM Applications With Prompt Engineering, issued November 2025.
- AWS Academy Graduate: AWS Academy Cloud Foundations, issued July 2025.
- Google Data Analytics: Foundations, Data, Data, Everywhere, issued April 2025 on Coursera.

ACHIEVEMENTS

- USAII Global AI Hackathon 2026: Finalist at Graduate Level, awarded by the United States Artificial Intelligence Institute for innovation, technical creativity, and applied AI on real world challenges.

LANGUAGES

English: Fluent, written and spoken
German: B1 in progress toward B2
Hindi: Native